

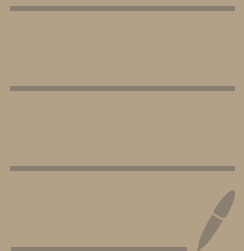
- ① Gronwall's inequality.
Lec 9. ② Continuous dependency
③ Comparison

• Book. Gerald Teschl.

① + ② Sec 2.4, Sec 2.6. or small Chapter 17.

③ Book: Rota. Sec. 1.11.

• This Lecture is more challenging.



(I) Gronwall's inequality. [T. Sec 2-4].

Let: Suppose $\psi(t)$ satisfies

$$\textcircled{1} \quad \psi(t) \leq \alpha(t) + \int_0^t \psi(s) \beta(s) ds, \quad \beta(s) \geq 0$$

- Then

$$\begin{aligned} \psi(t) &\leq \alpha(t) + \int_0^t \alpha(s) \beta(s) \exp\left(\int_s^t \beta(\tau) d\tau\right) ds \\ &= \alpha(0) \cdot \exp\left(\int_0^t \beta(s) ds\right) + \int_0^t \alpha'(s) \cdot \exp\left(\int_s^t \beta(\tau) d\tau\right) ds \end{aligned}$$

Integration by parts

Special case. $\alpha, \beta \equiv \text{constant}$

$$\psi(t) \leq \alpha \cdot e^{\beta t}$$

Pf: $A(t) = \int_0^t \psi(s) \beta(s) ds, \quad A(0) = 0$

$$\psi(t) = \frac{A'(t)}{\beta(t)}$$

we get $\textcircled{2} \Rightarrow$

$$\frac{A'(t)}{\beta(t)} \leq \alpha(t) + A(t)$$

$$\Rightarrow A'(t) \leq (\alpha\beta)(t) + \beta(t) A(t).$$

[this is also true if $\beta \geq 0$]

Solve ODE ineq, $\phi(s) = \exp\left(-\int_0^s \beta(\omega) d\omega\right)$

$$\Rightarrow \frac{d}{dt} [A \cdot \phi(t)] \leq \alpha(t) \phi(t)$$

$$\Rightarrow A(t) \phi(t) \leq \int_0^t \alpha(s) \beta(s) \phi(s) ds$$

$$\Rightarrow \psi(t) \leq \alpha(t) + A(t) \leq \alpha(t) + \int_0^t \alpha(s) \beta(s) \phi(s) \phi(t)^{-1} ds$$

. Give example:

$$\textcircled{1}. \quad \psi(t) \leq L \int_0^t \psi(s) ds + \alpha(0), \quad L > 0$$

Then $\psi \leq e^{Lt} \alpha(0) \dots$

$$\textcircled{2}. \quad \begin{cases} \frac{d}{dt} \psi(t) \leq L \psi(t) \\ \psi(0) = \psi_0 \end{cases}$$

$$\textcircled{3}. \quad \begin{cases} \frac{d}{dt} \psi(t) \leq L \psi(t) + M \\ \psi(0) = \psi_0 \end{cases}$$

(II)

Continuous
 $\frac{d}{dt} x = f(t, x)$
 $x(t_0) = x_0$

dependence.

$$\frac{d}{dt} y = g(t, y)$$
$$y(t_0) = y_0$$

• Suppose that f is locally Lipschitz

$$|f(t, x_1) - f(t, x_2)| \leq L \cdot |x_1 - x_2|$$

$$M = \max_{t, x} |f(t, x) - g(t, x)|$$

Can seen as integrate the error.

• Then.

$$|x(t) - y(t)| \leq |x_0 - y_0| \cdot e^{L|t-t_0|} + \frac{M}{L} \cdot (e^{L|t-t_0|} - 1).$$

• If initial perturbation and later perturbation are small.
 $\Rightarrow$ small error at later time

• pf: observe $|x(t) - y(t)| = \psi(t)$.

Integral form

$$x(t) - y(t) = x_0 - y_0 + \int_{t_0}^t [f(s, x(s)) - g(s, y(s))] ds$$

$$\begin{aligned} & |f(s, x(s)) - g(s, y(s))| \\ & \leq |f(s, x(s)) - f(s, y(s))| + |f(s, y(s)) - g(s, y(s))| \\ & \leq L |x(s) - y(s)| + M \end{aligned}$$

$$\Rightarrow |x(t) - y(t)| \leq |x_0 - y_0| + \int_{t_0}^t [L |x(s) - y(s)| + M] ds$$

Grönwall $\psi(t) \leq |x_0 - y_0| + M(t - t_0)$, $\beta \geq L$ Done.

$$\Rightarrow |x(t) - y(t)| \leq e^{L(t-t_0)} \|x_0 - y_0\| + \int_{t_0}^t \alpha'(s) \cdot e^{L(t-s)} ds$$

$$= e^{L(t-t_0)} \|x_0 - y_0\| + \frac{\alpha}{L} (e^{L(t-t_0)} - 1)$$

Cor: (Continuous dependence on initial data)

$$\begin{cases} \frac{d}{dt} x(t) = f(t, x(t)) \\ x(t_0) = x_0 \end{cases}$$

Solution given by $\Phi(t, x_0)$.

then Φ is Lipschitz in x_0 .

. Similarly. consider (Continuous dependence on parameter)

$$\begin{cases} \frac{d}{dt} x_a(t) = f(t, x_a(t), a) \\ x_a(t_0) = x_0 \end{cases}$$

where a is parameter. If $f(t, x, a)$ is Lipschitz in x, a .
Then solution $x_a(t)$ depends on a smoothly.

IV Comparison.

• Question: Can we get more quantitative control of ODE solution? *Gronwall ineq.*

• Consider

$$① \quad y'(t) \leq F(t, y(t))$$

• What can we say about f ?

Thm (Thm 7, Roach, Sec 1.11),

Let $F(t, y(t))$ satisfies Lipschitz cond for $t \geq a$.

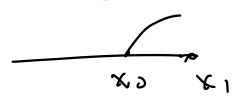
If f satisfies ① and g solves

$$g'(t) = F(t, g(t))$$

with $f(a) \leq g(a)$ then $f(t) \leq g(t), \forall t \geq a$.

pf: Argue by contradiction. Suppose $f(t_1) > g(t_1)$

and $t_0 = \max \{t \in [a, t_1] : f(t) \leq g(t)\}$. *since $f(a) \leq g(a)$ to exist.*

then $t_0 < t_1$. Define $d(t) = f(t) - g(t)$ 

$d(t_0) = 0, d(t) \geq 0, t \in [t_0, t_1]$
F is Lipschitz

$$d' = f' - g' \leq F(t, f(t)) - F(t, g(t))$$

$$\leq L |f(t) - g(t)| = L d(t), \quad t \in [t_0, t_1]$$

Gronwall $\Rightarrow d(t) \leq e^{L(t-t_0)} d(t_0) \Rightarrow d \geq 0$
on $[t_0, t_1]$

Contradiction !!

Below. Not covered at mid-term
(cover on Oct 30).

Applications: Thm

$$\begin{cases} \frac{d}{dt} x = f(t, x) \\ x(t_0) = x_0 \end{cases}$$

$$\begin{cases} \frac{d}{dt} y = f(t, y) \\ y(t_0) = y_0 \end{cases}$$

Ass 1: If $f(t, x) \leq f(t, y)$, $t \in [t_0, t_1]$

Ass 2: f or g is Lipschitz

Ass 3: $x_0 \leq y_0$.

$$x(t) \leq y(t), \quad \forall t \in [t_0, t_1] \quad \triangleq g(y)$$

• Ex. $\begin{cases} \frac{d}{dt} x = x + x^2 e^{-t} x + \frac{1}{(1+t^2)}, \\ x(t_0) = 1. \end{cases} \quad \frac{d}{dt} y = C \cdot y^n, \quad y(t_0) = 1$

• $f(x) \leq f(y)$. $\forall x \geq 1$. C large enough

• $x(t) \leq y(t)$

• $y(t)$ can be solved. explicitly